

It supports the end-to-end functionality required for infrastructure as a service, which includes compute, storage, network and user management; the RESTful API, the user interface and command-line tools.

CloudStack installation is done in two parts: 1) The management server; and 2) the cloud infrastructure. The latter is managed by a management server. CloudStack supports two types of networking—AWS-style networking and advanced networking for more sophisticated networks.

3.Eucalyptus

Eucalyptus is an open source + commercial private IaaS service provider. Eucalyptus is the short form for 'Elastic Computing Architecture for Linking Your Programs to Useful Systems'.

Main Features of Eucalyptus:

- Compatibility with Amazon Web Services API.
- Installation and deployment from source or DEB and RPM packages.
- Secure communication between internal processes via SOAP and WS-Security.
- Support for Linux and Windows virtual machines (VMs).
- Support for multiple clusters as a single cloud.
- Elastic IPs and security groups.
- Users and groups management.
- Accounting reports.
- Configurable scheduling policies and SLAs.

Each Eucalyptus cloud will have one or many Node Controllers per Cluster Controller. This controls VM activities—launching instances, inspection, termination of instances, and the clean-up. The Cluster Controller manages a collection of Node Controllers, manages access to public and private networks, controls execution of VMs, and manages virtual networking. Each Eucalyptus cloud will have multiple Cluster Controllers. A Cloud Controller manages virtualised resources; each Eucalyptus cloud will have a single CLC.

4.OpenNebula

OpenNebula manages virtual infrastructure to build private, public and hybrid infrastructure as Service Clouds. It orchestrates storage, network, virtualisation, monitoring and security. This open source project lets you build and manage virtualised enterprise data centres and IaaS clouds with core values such as openness, excellence, cooperation and innovation. OpenNebula was initiated as a research project in 2005 by Ignacio M Llorente and Ruben S Montero, and was released under the Apache licence.

OpenNebula provides AWS EC2 and EBS APIs, and the self-service portal for cloud consumers. It provides a powerful CLI that resembles typical UNIX commands. A catalogue of ready-to-run virtual appliances is available in the Appliance Marketplace. It provides support for hybrid cloud computing, with connectors for AWS. Security is available via user, group and role management,

as well as access control lists, auditing, and isolation at different levels. OpenNebula adheres to a modular and extensible architecture; thus, it is very easy to implement third-party tool integration. Customers get the best community support and SLA-based commercial support directly from the developers.

Distributed computing

5.Hadoop

Social networking (Twitter, Facebook, LinkedIn, etc), Web search engines (Google, Bing), and mobiles are some sources of big data. Scientists, analysts and architects deal with big data on a routine basis. The major part of this data has comprehensive and undiscovered relationships, which do not fit into traditional relational models.

Apache Hadoop is a software framework inspired by Google's MapReduce and Google File System (GFS) papers. Hadoop MapReduce is a programming model to write applications that rapidly process huge amounts of data in parallel, on large clusters of compute instances.

Hadoop can be used to analyse and process a variety of data to extract significant business operations intelligence, which remained hidden earlier. In normal scenarios, data moves to the computation node and then it is processed; but in Hadoop, data is processed where the data resides. The types of questions Hadoop helps answer are event analytics, large-scale Web click-stream analytics, revenue assurance and price optimisations, financial risk management and affinity engine, etc.

Amazon Elastic MapReduce is a Web service in the category of a public cloud. It enables researchers, analysts, developers and organisations to process vast amounts of data easily and cost-effectively. It utilises a hosted Hadoop framework running on the elastic infrastructure of Amazon Elastic Compute Cloud (Amazon EC2) and Amazon S3, and pre-configured EC2 instances (slave nodes) to distribute the MapReduce process.

6.Cassandra

Apache Cassandra is an open source distributed DBMS, which is a top-level project of the Apache Software Foundation. It is designed to handle very large amounts of data spread across many commodity servers. It provides a highly available service, which has architecture to prevent single point of failure. Apache Cassandra was developed at Facebook. In July 2008, it was distributed as an open source project on Google Code, while in March 2009, it was an Apache Incubator project and then released as a top-level project.

Cassandra provides a structured key-value store, where each node in the cluster has the same role. Data is distributed across the cluster, and each node can serve any request. It supports multi-data-centre replication for redundancy and disaster recovery. Data is automatically replicated to multiple nodes for fault-tolerance. It can be easily integrated with Hadoop with MapReduce support. Apache Pig and Apache Hive integration is also supported. There is also an SQL-like alternative called